

Danzel Yap

Project Engineer — LA Convention Center

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SUMMARY

Project Engineer on the \$2B Los Angeles Convention Center (LACC) expansion, managing structural foundation pours from layout through placement: RFIs, submittals, quantity takeoffs, budget tracking, and look-ahead scheduling. Built and deployed a suite of AI-powered construction management tools adopted by the project team, streamlining processes and saving the team hundreds of hours per week.

SKILLS

Project Engineering: RFIs, submittals, quantity takeoffs, concrete takeoffs, pour planning and logistics, embed coordination, budget and cost tracking, look-ahead scheduling, subcontractor coordination, MEP coordination, drawing review, field quality control, daily reporting, Revit.

Construction Software: Bluebeam Revu, Microsoft Excel, Microsoft Office, Google Workspace, Autodesk Revit.

Programming and AI: Python, JavaScript, C#, Java, MATLAB, VBA, LLM API integration (Anthropic, OpenAI, Gemini), Firebase, React, Google Apps Script.

Design: SolidWorks, Autodesk Fusion 360.

Languages: English (fluent), Mandarin (fluent), Cantonese (conversational).

EXPERIENCE

Webcor Craft

Project Engineer, Los Angeles, CA | [Month] 2025 - Present

- Manage structural foundation pours on the \$2B Los Angeles Convention Center (LACC) expansion, coordinating concrete operations, embed installation, quantity takeoffs, and pour logistics from layout through placement.
- Process RFIs and submittals, perform concrete quantity takeoffs, track budgets, and maintain look-ahead schedules across concurrent pour sequences.
- Coordinate installation and verification of 400 embeds across 75 embed types, tracked through a custom plan-map application built in-house.
- Designed, built, and deployed a suite of 6 AI-powered construction management applications adopted by the project management team, saving each project engineer approximately one day per week (detailed under Projects).
- Led evaluation of a third-party AI software vendor and recommended against adoption, saving the company \$75,000.

Clune Construction Company

Project Management Intern, San Francisco, CA | Jun 2024 - Aug 2024

- Reviewed construction drawings for internal office buildouts and ground-up infrastructure, leveled bids, basic MEP coordination, developed project budgets and schedules, researched practical applications of AI in construction.

Cal Poly Pomona Formula E

Nuts and Bolts Lead (School Project), Pomona, CA | Sep 2022 - Jun 2024

- Led design of suspension for a competition electric vehicle and provided photography services.

K-12 STEM Center USA

STEM Instructor (Part-Time), Claremont, CA | Oct 2022 - Mar 2026

- Taught robotics, Arduino, programming, and soldering to K-12 students; developed and launched an introductory AI course.

PROJECTS

AI in Construction Management Suite | 2024 - Present

- Built and deployed 6 web applications used daily by a live project team (pour tracking, embed tracking, tool inventory, crew attendance, scheduling, cylinder break tracking), plus an AI chat assistant integrated with Google Workspace and Discord.
- Spearheaded a research project on practical AI adoption in construction, interviewing 30+ industry professionals across California and Las Vegas — from senior VPs to interns — to map day-to-day workflows and where AI could deliver real value; findings formed the foundation of the suite.

FIRST Robotics Competition | California | Nov 2018 - Jun 2021

- Designed a wooden robotic arm for competition robots on a multidisciplinary team; secured team funding through grants and edited chairman presentation video.

EDUCATION

Bachelor of Science, Mechanical Engineering — California State Polytechnic University, Pomona | December 2025